

1.7: Hyperbolic Functions

1. Definitions

$$\cosh x = \frac{e^x + e^{-x}}{2}, \quad \sinh x = \frac{e^x - e^{-x}}{2}, \quad \tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}}.$$

Reciprocals: $\operatorname{sech} x = 1/\cosh x$, $\operatorname{cosech} x = 1/\sinh x$, $\operatorname{coth} x = 1/\tanh x$.

Key Values and Parity

- $\cosh 0 = 1$, $\sinh 0 = 0$, $\tanh 0 = 0$.
- $\cosh$ is **even**, $\sinh$ and $\tanh$ are **odd**.
- $\cosh x \geq 1$ for all real x ; range of $\tanh x$ is $(-1, 1)$.
- As $x \rightarrow \infty$: $\cosh x, \sinh x \rightarrow \frac{e^x}{2}$; $\tanh x \rightarrow 1$.

2. Fundamental Identities

$$\cosh^2 x - \sinh^2 x = 1$$

(Hyperbolic Pythagorean identity — the minus sign is the key difference from trig.)

$$1 - \tanh^2 x = \operatorname{sech}^2 x$$

$$\operatorname{coth}^2 x - 1 = \operatorname{cosech}^2 x$$

Addition Formulae

$$\sinh(A \pm B) = \sinh A \cosh B \pm \cosh A \sinh B$$

$$\cosh(A \pm B) = \cosh A \cosh B \pm \sinh A \sinh B$$

Double-Angle Formulae

$$\sinh 2x = 2 \sinh x \cosh x$$

$$\cosh 2x = \cosh^2 x + \sinh^2 x = 2 \cosh^2 x - 1 = 1 + 2 \sinh^2 x$$

Osborn's Rule

To convert a trig identity to its hyperbolic analogue: keep the form but change the sign of any term that is (or contains) a product of two sinhs.

e.g. $\cos 2A = 1 - 2 \sin^2 A$ becomes $\cosh 2A = 1 + 2 \sinh^2 A$ (sign flipped because of $\sin^2 A$).

3. Derivatives

$$\frac{d}{dx} \sinh x = \cosh x, \quad \frac{d}{dx} \cosh x = \sinh x$$

$$\frac{d}{dx} \tanh x = \operatorname{sech}^2 x$$

(The derivative of cosh is +sinh — no minus sign, unlike cos.)

4. Integrals

$$\int \cosh x \, dx = \sinh x + C, \quad \int \sinh x \, dx = \cosh x + C$$

$$\int \operatorname{sech}^2 x \, dx = \tanh x + C$$

For products like $\int \sinh^2 x \, dx$, use the double-angle identity:

$$\sinh^2 x = \frac{1}{2}(\cosh 2x - 1), \quad \cosh^2 x = \frac{1}{2}(\cosh 2x + 1).$$

5. Inverse Hyperbolic Functions

Logarithmic Forms

$$\operatorname{arsinh} x = \ln(x + \sqrt{x^2 + 1}), \quad x \in \mathbb{R}$$

$$\operatorname{arcosh} x = \ln(x + \sqrt{x^2 - 1}), \quad x \geq 1$$

$$\operatorname{artanh} x = \frac{1}{2} \ln\left(\frac{1+x}{1-x}\right), \quad |x| < 1$$

Derivation of arsinh

If $y = \operatorname{arsinh} x$, then $x = \sinh y = \frac{1}{2}(e^y - e^{-y})$. Let $u = e^y$: $u^2 - 2xu - 1 = 0 \Rightarrow u = x + \sqrt{x^2 + 1}$ (positive root). Take logs.

Derivatives of Inverses

$$\frac{d}{dx} \operatorname{arsinh} x = \frac{1}{\sqrt{x^2 + 1}}$$

$$\frac{d}{dx} \operatorname{arcosh} x = \frac{1}{\sqrt{x^2 - 1}}$$

$$\frac{d}{dx} \operatorname{artanh} x = \frac{1}{1 - x^2}$$

6. Standard Integrals (via Hyperbolic Substitutions)

Integrand	Substitution	Result
$\frac{1}{\sqrt{x^2 + a^2}}$	$x = a \sinh u$	$\operatorname{arsinh}(x/a) + C$
$\frac{1}{\sqrt{x^2 - a^2}}$	$x = a \cosh u$	$\operatorname{arcosh}(x/a) + C$ (for $x \geq a$)
$\sqrt{x^2 + a^2}$	$x = a \sinh u$, use $\cosh^2 u = \frac{1}{2}(\cosh 2u + 1)$	Quadrature formula

$$\int \frac{dx}{\sqrt{x^2 + 9}}$$

Let $x = 3 \sinh u$. Then $\sqrt{x^2 + 9} = 3 \cosh u$, $dx = 3 \cosh u \, du$.

Integrand becomes du , so integral $= u + C = \operatorname{arsinh}(x/3) + C$ (or equivalent log form).

7. Solving Equations with Hyperbolic Functions

Solve $2 \cosh x - \sinh x = 3$.

Sub $\cosh x = \frac{1}{2}(e^x + e^{-x})$, $\sinh x = \frac{1}{2}(e^x - e^{-x})$:

$$2 \cdot \frac{1}{2}(e^x + e^{-x}) - \frac{1}{2}(e^x - e^{-x}) = 3$$

$$\frac{1}{2}e^x + \frac{3}{2}e^{-x} = 3. \text{ Multiply by } 2e^x: e^{2x} + 3 = 6e^x \Rightarrow e^{2x} - 6e^x + 3 = 0.$$

$$e^x = 3 \pm \sqrt{6}. \text{ Both positive, so } x = \ln(3 \pm \sqrt{6}).$$

8. Exam Style Strategy

- Convert to exponentials early for solving equations.
- Use Osborn's rule to translate trig identities — but always double-check by direct expansion.
- Hyperbolic substitutions often make "root of $x^2 + a^2$ " integrals cleaner than trig substitutions.
- Know when arcosh gives $\pm$ — the logarithmic form picks the positive branch; think about whether both signs are wanted.

